On Simplification, we obtain $\overrightarrow{BC}\times\overrightarrow{CA} = \overrightarrow{CA}\times\overrightarrow{AB}$ ... (2)

Equations (1) and (2), we get

$$\overrightarrow{AB}\times\overrightarrow{BC} = \overrightarrow{CA}\times\overrightarrow{AB} = \overrightarrow{BC}\times\overrightarrow{CA}.$$

So, $|\overrightarrow{AB}\times\overrightarrow{BC}| = |\overrightarrow{CA}\times\overrightarrow{AB}| = |\overrightarrow{BC}\times\overrightarrow{CA}|$. Then, we get

$$ca\sin(\pi - B) = bc\sin(\pi - A) = ab\sin(\pi - C).$$

That is, $ca\sin B = bc\sin A = ab\sin C$. Dividing by abc, leads to

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} \quad \text{or} \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Example 6.5

Prove by vector method that $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$.

Solution

Let $\hat{a} = \overrightarrow{OA}$ and $\vec{b} = \overrightarrow{OB}$ be the unit vectors making angles α and β respectively, with positive x-axis, where A and B are as shown in the Fig. 6.10. Then, we get $\hat{a} = \cos\alpha\hat{i} + \sin\alpha\hat{j}$ and $\hat{b} = \cos\beta\hat{i} + \sin\beta\hat{j}$,

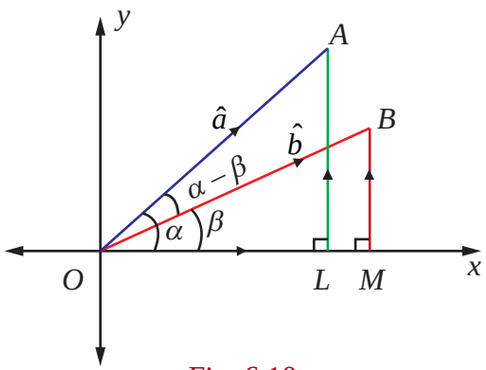


Fig. 6.10

The angle between $\hat{a}$ and $\hat{b}$ is $\alpha - \beta$ and, the vectors $\hat{b}, \hat{a}, \hat{k}$ form a right-handed system.

Hence, we get

$$\hat{b}\times\hat{a} = |\hat{b}||\hat{a}|\sin(\alpha - \beta)\hat{k} = \sin(\alpha - \beta)\hat{k}. \quad \text{... (1)}$$

On the other hand,

$$\hat{b}\times\hat{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\beta & \sin\beta & 0 \\ \cos\alpha & \sin\alpha & 0 \end{vmatrix} = (\sin\alpha\cos\beta - \cos\alpha\sin\beta)\hat{k} \quad \text{... (2)}$$

Hence, equations (1) and (2), leads to

$$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta.$$

6.3.3 Application of dot and cross products in Geometry

Example 6.6 (Apollonius's theorem)

If D is the midpoint of the side BC of a triangle ABC, show by vector method that $|\overrightarrow{AB}|^2 + |\overrightarrow{AC}|^2 = 2(|\overrightarrow{AD}|^2 + |\overrightarrow{BD}|^2)$.

Solution

Let A be the origin, $\vec{b}$ be the position vector of B and $\vec{c}$ be the position vector of C. Now D is the midpoint of BC, and so the, position vector of D is $\frac{\vec{b} + \vec{c}}{2}$. Therefore, we have

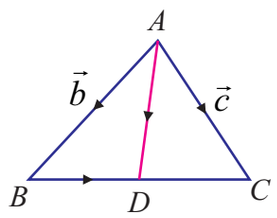


Fig. 6.11